Lab 8: BJTs

ECE 0102: Microelectronic Circuits + Lab

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Team Roles:

Activity	**Student Name**
Circuit Construction	Stefan Zhang
Data Collection	Knight Liu
Data Analysis	Stefan Zhang; Knight Liu
Answers to Questions	Stefan Zhang; Knight Liu
Lab Report Writing	Stefan Zhang

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1) Objectives and Background

The objective of this lab is to measure the current-voltage characteristics of a 2N2222A NPN bipolar junction transistor and to develop a simple model for its behavior. Two main parameters were extracted in this experiment: the saturation current I_S and the common-emitter current gain β.

For an NPN transistor operating in the forward-active region, the collector current can be modeled by

$$I_C = I_S e^{V_{BE}/V_T}$$

where V_T is the thermal voltage. At room temperature,

$$V_T \approx 25\,\mathrm{mV}$$

The common-emitter current gain is defined as

$$\beta = \frac{I_C}{I_B}$$

2) Equipment Used

Table 1: Equipment used in Lab 8

Item	**Quantity**
2N2222A NPN transistor	1
Resistor, $100\,\mathrm{k\Omega}$	1
Resistor, $1\,\mathrm{k\Omega}$	1
DC Power Supply	1
Digital Multimeter	1
Breadboard and wires	1 set

3) Action 1: Measurement of I_C and Extraction of I_S

Question

Set $V_{IN} = 1\,\mathrm{V}$ and measure I_C at $V_{CE} = 1$ to $5\,\mathrm{V}$. Then develop a model by finding the saturation current I_S using

$$I_C = I_S e^{V_{BE}/V_T}.$$

Setup

The circuit used a $100\,\mathrm{k\Omega}$ base resistor, a $1\,\mathrm{k\Omega}$ collector resistor, and a 2N2222A NPN transistor. The emitter was connected to ground. The input voltage was fixed at $V_{IN} = 1\,\mathrm{V}$, and V_{CC} was adjusted to obtain different V_{CE} values.

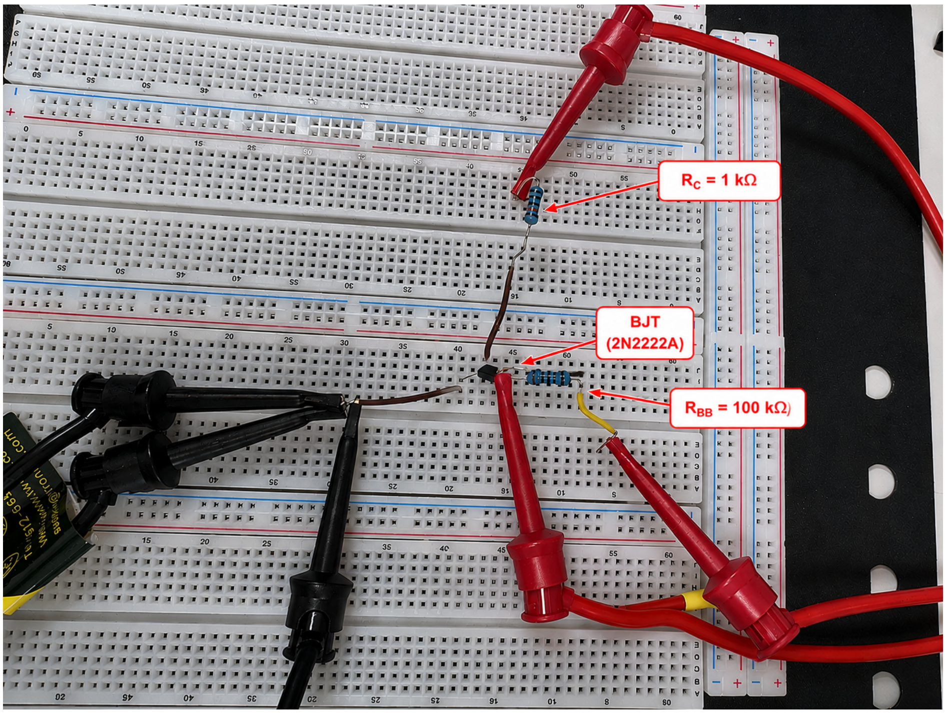

Figure 1: Annotated experimental setup for Action 1

Data and Calculation

The thermal voltage was taken as

$$V_T = 25\,\mathrm{mV} = 0.025\,\mathrm{V}.$$

The saturation current was calculated by

$$I_S = I_C e^{-V_{BE}/V_T}.$$

Table 2: Measured and calculated collector current in Action 1

V_{CC} [V]	V_{CE} [V]	V_{BE} [V]	I_C from Action #1.1 [A]	I_C from Action #1.2 [A]	Error [%]
1.98	1.0	0.609	9.97×10^{-4}	1.027×10^{-3}	3.01
2.99	2.0	0.609	1.004×10^{-3}	1.027×10^{-3}	2.30
4.00	3.0	0.609	1.010×10^{-3}	1.027×10^{-3}	1.69
5.00	4.0	0.608	1.017×10^{-3}	9.868×10^{-4}	2.97
6.01	5.0	0.608	1.024×10^{-3}	9.868×10^{-4}	3.63
Average Error					2.72

The extracted saturation current was

Table 3: Extracted saturation current

Parameter	Value
I_S (A)	2.70×10^{-14}

Answer

The measured V_{BE} values were about 0.608–0.609 V. Using the forward-active BJT equation, I_S was calculated from each measured point and then averaged. The final model used

$$I_C = (2.70 \times 10^{-14})e^{V_{BE}/0.025}.$$

The average error between the measured current and the model current was 2.72%, which shows that the exponential model fits the measured data well.

4) Action 2: Measurement of β

Question

Set $V_C = 3\,\text{V}$ and measure I_C at $I_B = 5$ to $40\,\mu\text{A}$. Then create a plot of β versus I_B and check whether the measured $\beta = h_{FE}$ is reasonable.

Data and Calculation

The common-emitter current gain was calculated by

$$\beta = \frac{I_C}{I_B}.$$

Table 4: Measured collector current and calculated current gain

V_{IN} **[V]**	I_B **[μA]**	I_C **[A]**	$\beta = I_C/I_B$
1.12	5	1.330×10^{-3}	266.0
1.63	10	2.610×10^{-3}	261.0
2.15	15	3.939×10^{-3}	262.6
2.66	20	5.269×10^{-3}	263.5
3.16	25	6.585×10^{-3}	263.4
3.67	30	7.928×10^{-3}	264.3
4.18	35	9.310×10^{-3}	266.0
4.68	40	1.0666×10^{-2}	266.7
Average β			264.2

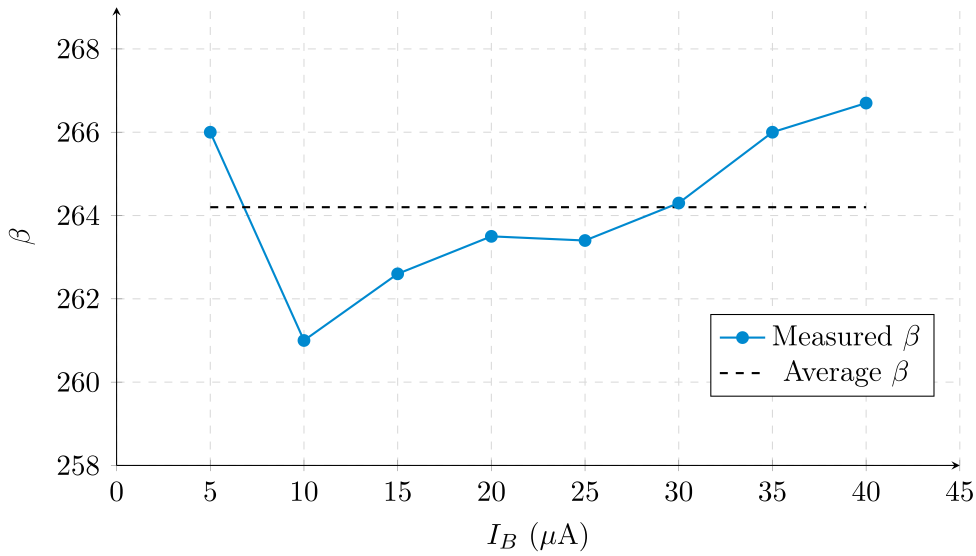

Figure 2: Common-emitter current gain β versus base current I_B

Question: Check the datasheet and explain if your (=hFE) is reasonable

Answer

The measured β ranged from 261.0 to 266.7, and the average value was

$$\boxed{\beta_{\text{avg}} = 264.2}$$

Since β is the same parameter as the datasheet DC current gain h_{FE}, this value is reasonable for a 2N2222A transistor. The small variation also indicates that the transistor stayed in the forward-active region during the measurement. According to the 2N2222A datasheet, h_{FE} is usually in the range of tens to several hundreds depending on I_C and V_{CE}. Since the measured average value is $\beta_{\text{avg}} = 264.2$, it is within a reasonable range.

5) Conclusion

In this lab, the saturation current and current gain of a 2N2222A NPN transistor were measured. The extracted saturation current was

$$I_S = 2.70 \times 10^{-14}\,\text{A}$$

and the average current gain was

$$\beta_{\text{avg}} = 264.2.$$

The model current matched the measured current with an average error of 2.72%, so the experimental results are consistent with the forward-active BJT model.

References

[1] SCUPI 2026S Microelectronics Circuit Lab Manual, Lab #8: BJTs.

[2] Sedra, A. S., Smith, K. C., *Microelectronic Circuits*, Oxford University Press.

[3] Central Semiconductor Corp., *2N2222A NPN Silicon Transistor Datasheet.*